

Q. 1 (Event gymnastics). Let E_1, E_2, E_3 be three events. Express the following “English language” events mathematically in terms of set operations ($\cap, \cup, \setminus, ^c$):

- a. Both E_1 and E_2 occur, but E_3 does not.
- b. Exactly two of the three events occur.
- c. At least one of the three events occurs.
- d. None of the three events occur.

The symmetric difference of two sets A, B is defined as $A \triangle B := (A \setminus B) \cup (B \setminus A)$.

- e. What event does $E_1 \triangle E_2$ represent in English? (It can be helpful to draw a picture.)

Q. 2 (Coin flips). Let us create a probability model of 3 coin flips by assuming that each possible sequence of heads/tails is equally likely to occur.

- a. What is a good sample space for this model? How can you define the probability measure on this sample space?
- b. What is the event that the first coin comes up heads? That the second coin comes up tails? What are the probabilities of these events?
- c. What is the probability that an odd number of coins come up heads?

Now consider 100 coin flips. Again, we introduce the modelling assumption that each possible sequence of heads/tails is equally likely to occur.

- d. What is a good sample space for this model? What is the event that the n th coin comes up heads? How can you define the probability measure on this sample space?
- e. What is the probability that exactly one coin comes up heads?

Now consider an *infinite* sequence of coin flips.

- f. What is a good sample space for this model?
- g. What problem arises when you try to define a probability on this space such that every sequence of coin flips is equally likely, as in the previous parts of this problem? Do you know another example where such a “paradox” arises?

(We will discuss sequences of coin flips in much more detail in a few lectures after we have learned some additional concepts.)

Q. 3 (Darts). In this exercise, we are going to try to model a random point in the disc

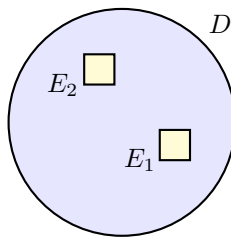
$$D := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

You could view this as a model of the point at which a dart hits the dartboard.

a. What is a natural sample space for this problem?

We are going to try to create a model where every part of the dartboard is “equally likely” to be hit. (This may not be a good description for darts players of your esteemed caliber, but it should be more than adequate for my negligible skills.)

How can we make this precise? One natural idea is as follows. If each point of the dartboard is equally likely to be hit, then the probability of hitting a set of the same shape should not change if we move it around the dartboard. For example, we should have the same probability of hitting the set E_1 and of hitting the set E_2 in this picture:



b. The above picture suggests that we might choose the probability of hitting a subset $A \subseteq D$ of the dartboard to be proportional to the area of this set. If we follow this idea, how can we define the probability measure on our sample space?

c. In this model, what is the probability that the dart lands in the ring

$$R := \{(x, y) \in \mathbb{R}^2 : \frac{1}{2} < x^2 + y^2 < \frac{3}{4}\} \quad ?$$

It is not at all obvious that the above model for throwing a dart is the “right” model. Here is an equally plausible way of modelling (at least in absence of additional information about the darts player). Think of a point on the dartboard in polar coordinates (r, ϕ) : that is, $r \in [0, 1]$ is its distance from the center of the dartboard and $\phi \in [0, 2\pi]$ is the angle. It seems reasonable to assume that each angle and each distance from the center will be equally likely. In particular, this suggests that the probability that the distance r from the center falls in an interval $B \subseteq [0, 1]$ should be proportional to the length of the interval (by the same reasoning as in part b.), and similarly for the angle.

d. In this model, what is the probability that the dart lands in the ring R ?

e. Compare the answers to parts c. and d. Explain why they are the same/different.

As you can see, in general our intuitive notions of “random” and “equally likely” are far from obvious. You must always be careful when modelling probabilities!